

Lab : Running a Vulnerability Scan

Estimated duration: 0 minutes

Overview:

In this lab, you will use OpenVAS to run a basic unauthenticated vulnerability scan on a deliberately vulnerable VM such as Metasploitable2. You will then review and analyze the findings using the Greenbone Security Assistant (GSA) web interface.

This activity provides a foundational hands-on experience in vulnerability scanning, an essential task in both offensive and defensive cybersecurity roles.

Learning objectives

After completing this lab, you will be able to:

- Launch an unauthenticated vulnerability scan using the OpenVAS
- Monitor scan progress and interpret task logs
- Analyze vulnerability reports, including CVEs and risk levels
- Recognize default misconfigurations and exploitable weaknesses

Prepare the vulnerable target: Metasploitable 2

Before running a vulnerability scan, you need a target system. In this section, you will download and configure Metasploitable, a purposely vulnerable Linux VM designed for penetration testing and security research.

Run Metasploitable2 on your physical computer (host), not inside another VM. Nested virtualization can cause network problems and failures.

Step 1: Download Metasploitable 2

1. Navigate to the official download link:

<https://sourcefire.com/resources/patches/metasploitable2/>

2. Download the .iso file

sourceforge.net/projects/metasploitable/files/Metasploitable2/

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Metasploitable Files

Metasploitable is an intentionally vulnerable Linux virtual machine
Brought to you by: [rapid7user](#)

Summary **Files** Reviews

 **Download Latest Version**
metasploitable-linux-2.0.0.zip (865.1 MB)

 Get an email when there's a new version of Metasploitable

Enter your email address

Home / Metasploitable2

Name	Modified	Size
 Parent folder		
metasploitable-linux-2.0.0.zip	2019-08-19	865.1 MB

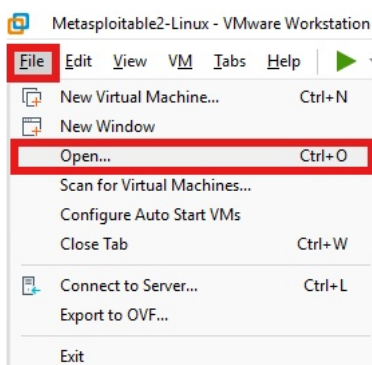
Once downloaded, extract the archive.

Step 2: Set up the VM

1. Start the virtual machine:

on a workstation/ lab:

- Open VMware Workstation or Player
- Click File > Open...
- Browse to and select the extracted file (usually Metasploitable.ova).
- Follow any prompts to finish importing.

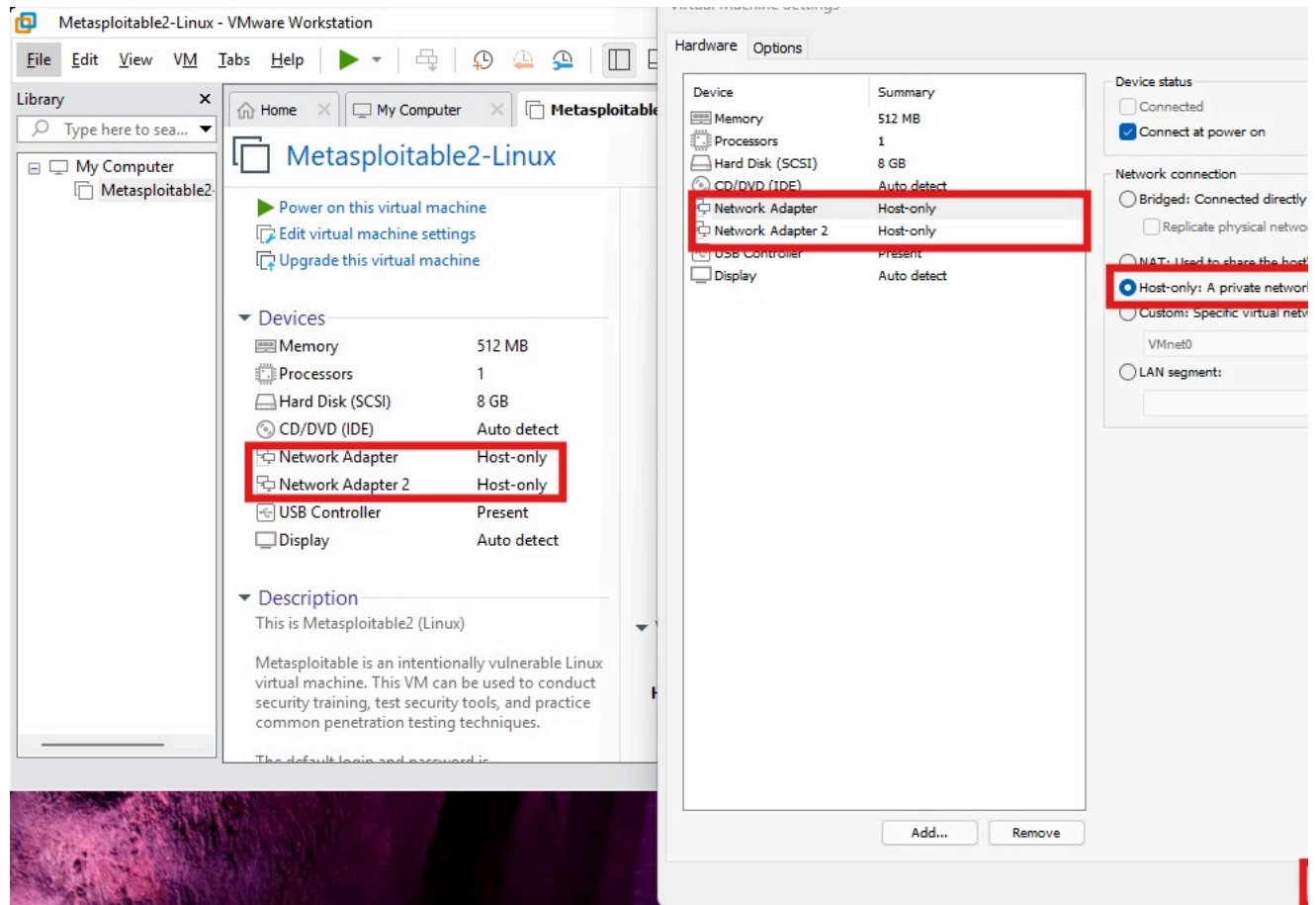


2. Configure network settings:

- Set the network adapter to:

- NAT (recommended for isolated lab).

Change all adapters to this setting to ensure safe usage and not have the VM open to attack from the internet side.



Step : oot and log in

1. Start the VM

2. When prompted, log in with the default credentials:

- o **serna** **emsfa** min
- o **ass** **rd** min

Run the following command to identify the IP address:

```
ifconfig
```

- o Note the IP address usually from eth0 (for example, . .) you will need it for the OpenVAS scan.

```

msfadmin@metasploitable:~$ ifconfig
eth0: Link encap:Ethernet HWaddr 00:0c:29:01:a0:09
      inet addr:192.168.183.128 Bcast:192.168.183.255 Mask:255.255.255.0
      inet6 addr: fe80::20c:29ff:fe01:a009/64 Scope:Link
      UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
      RX packets:6 errors:0 dropped:0 overruns:0 frame:0
      TX packets:48 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:1000
      RX bytes:926 (926.0 B) TX bytes:5352 (5.2 KB)
      Interrupt:17 Base address:0x2000

lo: Link encap:Local Loopback
      inet addr:127.0.0.1 Mask:255.0.0.0
      inet6 addr: ::1/128 Scope:Host
      UP LOOPBACK RUNNING MTU:16436 Metric:1
      RX packets:96 errors:0 dropped:0 overruns:0 frame:0
      TX packets:96 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:0
      RX bytes:21437 (20.9 KB) TX bytes:21437 (20.9 KB)

msfadmin@metasploitable:~$ _

```

Important security note

- Metasploitable 2 is extremely insecure by design and should only run in isolated environments (host-only or internal networks).
- Always turn it off after lab use.

What it is

Metasploitable 2 simulates real-world systems with unpatched services and weak configurations. It is used widely by ethical hackers and researchers to practice vulnerability scanning, exploitation, and risk analysis without breaking any laws or harming real systems.

Conduct a vulnerability scan with OpenVAS in Kali Linux

Step 1: Start OpenVAS services

Ensure that OpenVAS is operational and its core components are running.

Run the following commands:

```

su -c 'systemctl start openvas'
su -c 'systemctl enable openvas'

```

Confirm the following are running:

- greenbone-vulnerability-manager (greenbone Vulnerability Manager)
- openvas-scanner
- gsa (greenbone Security Assistant Daemon)

Confirm Metasploitable2 can be seen by Kali Linux:

```

kali@kali: ~
File Actions Edit View Help

(kali@kali)-[~]
$ ping 192.168.183.128
PING 192.168.183.128 (192.168.183.128) 56(84) bytes of data.
64 bytes from 192.168.183.128: icmp_seq=1 ttl=63 time=0.885 ms
64 bytes from 192.168.183.128: icmp_seq=2 ttl=63 time=0.729 ms
64 bytes from 192.168.183.128: icmp_seq=3 ttl=63 time=0.644 ms
64 bytes from 192.168.183.128: icmp_seq=4 ttl=63 time=1.16 ms
64 bytes from 192.168.183.128: icmp_seq=5 ttl=63 time=0.840 ms
^C
--- 192.168.183.128 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4068ms
rtt min/avg/max/mdev = 0.644/0.851/1.159/0.175 ms

(kali@kali)-[~]
$

```

- run a ping from a terminal to the metasploitable2 IP.

is still not be a local host

If ping does not work, check that both VMs are on the same Host-Only Network and are powered on. You may need to restart one or both VMs. Always use the Metasploitable2 IP (usually starting with 10.10.10.x or 10.10.10.x) as shown in the figure.

Using 127.0.0.1 (localhost) or the Kali VM IP will not scan the correct target.

Why this matters

These services form the backbone of OpenVAS. Without them running properly, scans cannot be performed, and results cannot be accessed through the web UI. Also, if Kali cannot see the metasploitable2 VM, the scans will not have the needed results to move forward.

Step 2: Log into the web interface

Access the OpenVAS web-based interface to figure and run scans.

1. Open Firefox (or another browser) in your Kali
2. Navigate to:

https://10.10.10.2:5443

Log in using the admin credentials you created during setup. You may need to accept the self-signed certificate warning.

Add the Target in OpenVAS
Go to Configuration -> Targets in the OpenVAS web interface (SA).
Create a new target:
Name Metasploitable2
Hosts Enter the Metasploitable IP
Port List All IANA assigned TCP
Click on save

Why this matters

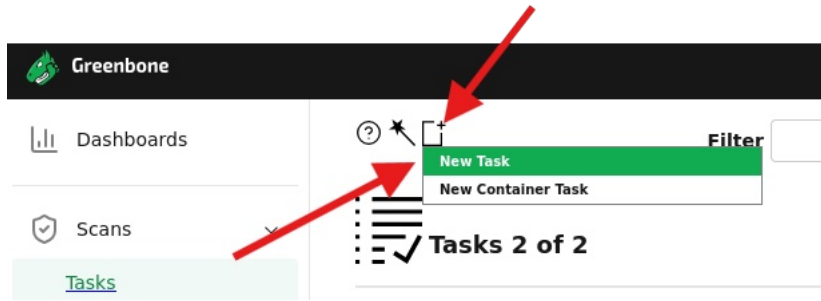
This interface is where analysts manage scans, view reports, and figure scan policies.

Step 3: Launch a quick scan

Run a simple unauthenticated scan against a known vulnerable target

1. From the side pannel, select Scans/Targets and then click:

Scans



2. Enter:

- o a e.g., ab can Metasploitable2
- o Scan targets Metasploitable2

Use default settings:

- o Profile: full scan fast

Click create and then in start to start the scan.

Name ↑	Status ↑↓	Reports ↑↓	Last Report ↑↓	Severity ↑↓	Tr
Lab Scan Metasploitable2	0 %	1			
New Quick Task (Automatically generated by wizard)	Done	1	Mon, Jun 16, 2025 12:29 PM Coordinated Universal Time	5.9 (Medium)	

Scan time: May take 10- 0 minutes depending on system resources, if your scan shows no vulnerabilities or a severity of 0, double-check that the correct Metasploitable2 IP was used and that the VM is running, make sure the metasploitable2 IP can be pinged from the Kali Linux VM. Use a terminal and run a ping command from the Kali VM to confirm this.

ping Metasploitable2 address

Step : Monitor scan progress

ur ~~Track~~ the scan s execution status and timing.

1. o to:

cans as s

2. Locate your scan in the list.

Observe:

- r es bar



Name ↑	Status ↑↓	Reports ↑↓
Lab Scan Metasploitable2	94 %	1

- Last re rt ti e

y t is atters:

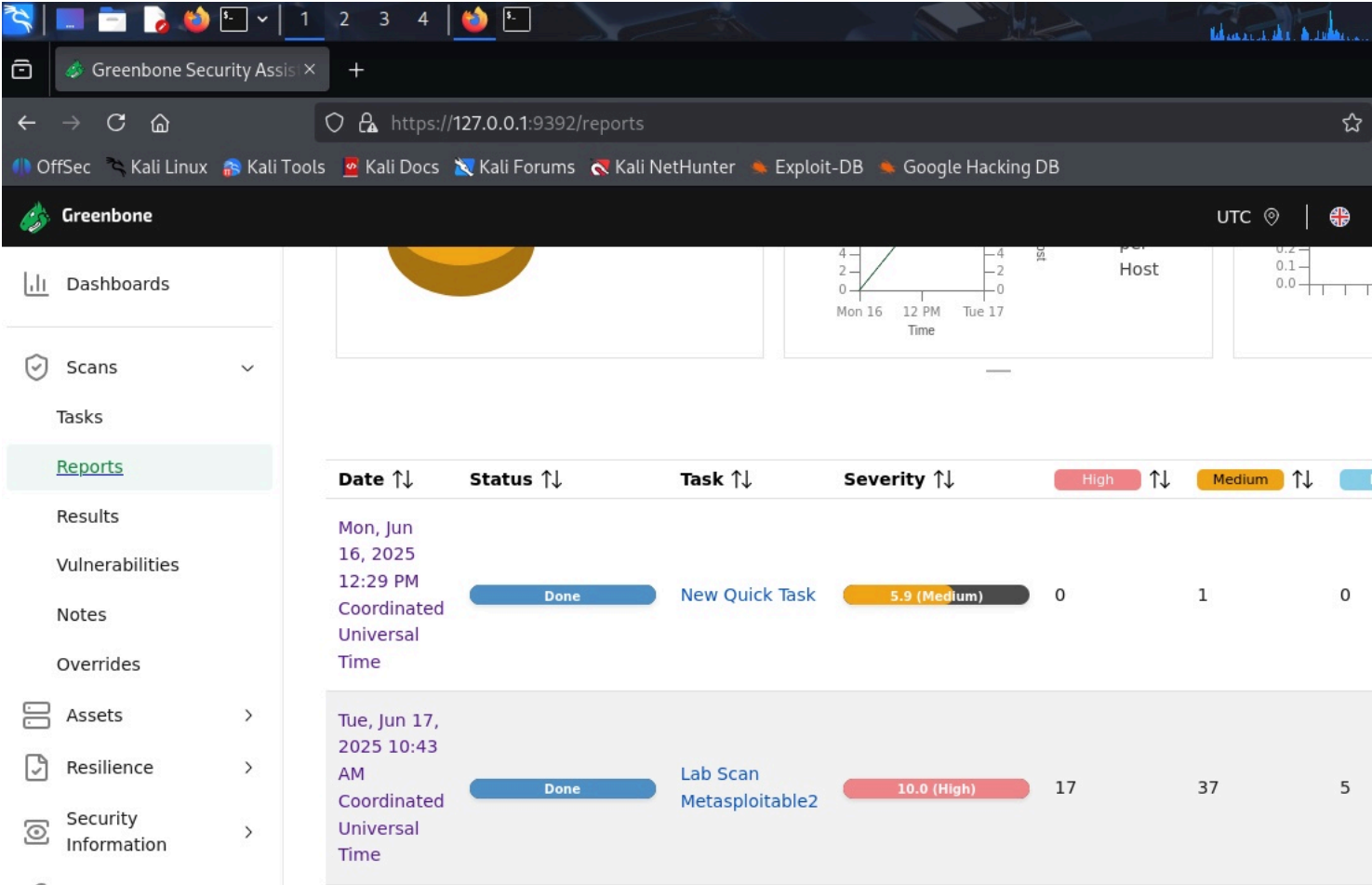
Understanding scan duration and status helps security teams plan and assess operational timelines during assessments.

Step : Analy e the report

ur ~~Inter~~pret scan results, assess vulnerabilities, and understand their implications.

1. o to:

cans eports



2. Open the latest report and review:

- Severity list
- Detected ports and services
- Vulnerabilities by VSS score

Click individual results to view:

- CVE IDs
- Exploitation details (if available)
- Remediation steps

Why this matters

Security reports must be read critically. VSS scores give context to risk, while CVE data ties findings to known exploits.

Step 1: identify sample findings

Here are common examples of what you might find in your scan:

type	sample
Open Port	Port 21 (FTP)
Missing Patch	Apache 2.2. DoS vulnerability
Default Creds	ms-admin ms-admin (common default)
Critical CVE	CVE-2017- Apache Struts RCE

1. Document at least three unique vulnerabilities found in your own report.

Challenge: Defend against the findings

Blue team report:

Based on your scan report, identify one HIGH or CRITICAL severity vulnerability and answer:

- # Skills Network